1b) 123

$\boxed{0111\ 1011}$

$123 \div 2$
$= 61 \quad R\ 1$
$61 \div 2$
$= 30 \quad R\ 1$
$30 \div 2$
$= 15 \quad R\ 0$
$15 \div 2$
$= 7 \quad R\ 1$
$7 \div 2$
$= 3 \quad R\ 1$
$3 \div 2$
$= 1 \quad R\ 1$
$1 \div 2$
$= 0 \quad R\ 1$

2b) 1000001

$= 1 \cdot 2^0 + 1 \cdot 2^6$

$\boxed{= 65_{10}}$

3b) 65

$= 65 \div 16$

$= 4 \quad R\ 1$

$= 0 \quad R\ 4$

$\boxed{= 41_{16}}$

4d) 2BH

$\boxed{= 0010\ 1011}$

to decimal

$= 1 \cdot 2^0 + 1 \cdot 2^1 + 1 \cdot 2^3 + 1 \cdot 2^5$

$\boxed{43}$

21 c) 1 word = 16 bits

$\boxed{\text{4 words in 64 bits}}$

22) giga = 10^9

$\frac{1 \text{ page}}{2000 \text{ bytes}}$ $\frac{2 \cdot 10^9 \text{ bytes}}{2 \text{ gigabytes}}$

$\boxed{1000\,000 \text{ pages}}$

26) Memory = 2^n

$= 2^{16} = 65536$

$\frac{65536 \text{ bytes}}{1024 \text{ bytes}}$ $\frac{1 \text{ kilobyte}}{}$

$\boxed{= 64 \text{ kilobytes}}$

27) The address bus is unidirectional because address is sent to memory from external devices but not in reverse direction.

The data bus is bidirectional since data is transferred from memory to external devices as well as from external devices to memory.